

YAMAHA NETWORK BOARD

EtherCAT

User's Manual

RCX340

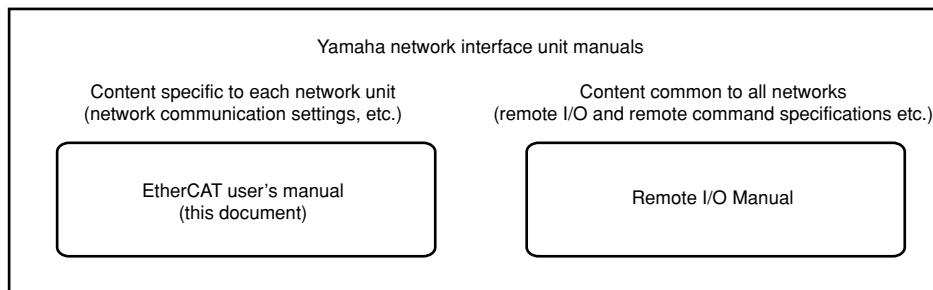
Important information before reading this manual	1
Introduction	1
Safety Precautions (Always read before starting use)	2
Warranty	4
1. Overview of functions	5
2. Controller status transitions	6
3. Input/output assignments	7
4. Part names and functions	9
5. Flow until operation begins	10
6. Connection	11
6.1 Network structure	11
6.2 Cabling	12
6.3 Noise countermeasures	12
7. ESI Files	13
8. Parameter Settings	14
9. Monitoring communication data	15
9.1 Using the programming box	15
9.1.1 Switching the port number	16
9.1.2 Input/output list display	17
9.1.3 Input/output details display	17
9.1.4 Switching the output status	18
9.2 Using RCX-Studio PC support software	19
10. Disabling an option unit	22
10.1 Making settings	23
10.1.1 Using the programming box	23
10.1.2 Using RCX-Studio PC support software	24

11. Actions to take if communication is not established	26
11.1 Reconsider the startup steps	26
11.2 Viewing the alarm codes in the 7-segment LED	27
11.3 Diagnostics by alarm message	27
11.3.1 Alarms related to network unit connections and settings	27
11.3.2 Alarms related to I/O on/off status	28
11.3.3 Alarms that the customer cannot resolve	28
11.3.4 Checking alarm codes in the programming box	29
11.3.5 Checking alarm codes in RCX-Studio PC support software	29
12. Specifications	30
12.1 Network specifications	30
12.2 Input/output specifications	31
13. Reserved word list	32

Important information before reading this manual

Introduction

The Yamaha network interface unit documentation consists of a User's Manual that is specific to each network unit that describes the specifications and communication settings of that unit, and a Remote I/O Manual describing the remote I/O specifications that are common to all network units.



For details on network unit specifications and communication settings, refer to this User's Manual.

For details on communication functions between the controller and the host control device, such as remote I/O and remote commands, refer to the Remote I/O Manual.

For details on the functions of the robot controller unit, refer to the following manuals.

RCX340 Operator's Manual..... for operation of the controller unit

RCX340 User's Manual for specifications and settings of the controller unit

RCX340 Programming Manual for the programming language used by Yamaha robot controllers

Safety Precautions (Always read before starting use)

Before using this product, be sure to read this manual carefully as well as the robot controller user's manual and programming manual. Take sufficient precautions to ensure safety and handle the product correctly. The cautions given in this manual are related to this product. Refer to the robot controller user's manual for details on the cautions to be taken with the robot controller system using this product. The safety precautions are ranked as "WARNING" and "CAUTION" in this manual.



WARNING

Failure to follow WARNING instructions could result in serious injury or death to the operator or person servicing the product.



CAUTION

Failure to follow CAUTION instructions may result in injury to the operator or person servicing product, or damage to the product or peripheral equipment.



NOTE

This indicates a supplementary explanation in the operation.

Note that some items described as "CAUTION" may lead to serious results depending on the situation. In any case, important information that must be observed is explained.

Store this manual where it can be easily referred to, and make sure that it is delivered to the end user.

EtherCAT is a registered trademark and patented technology licensed from Germany Beckhoff Automation GmbH.

■ Precautions for design



WARNING

- For information about the status of the network system and robot controller in the event that a communication problem occurs in the network system, refer to the manual of the host control device as well as to this document. Configure an interlock circuit in the sequence program so that the system, including the robot controller, will work safely WITH using the communication status information.
- The SAFETY connector has an emergency stop terminal to trigger emergency stop of the robot controller. By using this terminal, prepare a physical interlock circuit so that the system including the robot controller will work safely.



CAUTION

The control line and communication cable must not be bound with or placed near the main circuit or power line. Separate these by at least 100mm. Failure to observe this could lead to communication error or declining throughput caused by noise.

■ Precautions for installation



WARNING

- Always crimp, press-fit or solder the connector wiring with the maker-designated tool, and securely fix the connector to the module.
- Always shut off all phases of the power supply externally before starting installation or wiring work. Failure to shut off all phases could lead to electric shocks or product damage.



CAUTION

- Use the robot controller in the environmental conditions specified in this manual. Operation outside the specified environmental range may cause electric shocks, fire, malfunction or product damage or deterioration.
- Do not directly touch the conductive portions or electronic components of a network module.
- Never directly touch the controller's interior areas.
- Accurately connect each cable connector to the mounting section.
Failure to observe this could lead to malfunctions caused by a connection fault.



WARNING

Always shut off all phases of the power supply externally before starting installation or wiring work. Failure to shut off all phases could lead to electric shocks or product damage.



CAUTION

- Make sure that foreign matter, such as cutting chips or wire scraps, does not enter the robot controller.
- Communication cables that contact network modules must be kept inside a duct or secured by clamps. Failure to place the cable in a duct or secure it by a clamp could damage the cable or module by shifting, movement or unintentional pulling the cable, or cause malfunction by poor contact condition.
- When disconnecting a connector from the network module, grasp the connector rather than pulling on the cable. Pulling on the cable could damage the cable and module, possibly causing a poor contact condition which could result in malfunctions.

■ Precautions for starting and maintenance



WARNING

- Do not touch the terminals while the power is ON. Failure to observe this could lead to malfunctions.
- Always shut off all phases of the power supply externally before performing cleaning or wiring work. Failure to shut off all phases could lead to electric shocks, product damage or malfunctions.
- Never disassemble or modify any of the robot controller modules.
Failure to observe this could lead to trouble, malfunctions, injuries or fires.



CAUTION

Power must be supplied to the robot controller only after supplying power to the host control device. The robot controller will enter an error state if communication is not established within a certain length of time after the controller starts.

■ Precautions for disposal



CAUTION

Dispose of this product as industrial waste.

Warranty

For information on the warranty period and terms, please contact our distributor where you purchased the product.

■ This warranty does not cover any failure caused by:

1. Installation, wiring, connection to other control devices, operating methods, inspection or maintenance that does not comply with industry standards or instructions specified in the YAMAHA manual;
2. Usage that exceeded the specifications or standard performance shown in the YAMAHA manual;
3. Product usage other than intended by YAMAHA;
4. Storage, operating conditions and utilities that are outside the range specified in the manual;
5. Damage due to improper shipping or shipping methods;
6. Accident or collision damage;
7. Installation of other than genuine YAMAHA parts and/or accessories;
8. Modification to original parts or modifications not conforming to standard specifications designated by YAMAHA, including customizing performed by YAMAHA in compliance with distributor or customer requests;
9. Pollution, salt damage, condensation;
10. Fires or natural disasters such as earthquakes, tsunamis, lightning strikes, wind and flood damage, etc;
11. Breakdown due to causes other than the above that are not the fault or responsibility of YAMAHA;

■ The following cases are not covered under the warranty:

1. Products whose serial number or production date (month & year) cannot be verified.
2. Changes in software or internal data such as programs, points, calibration, or registered models that were created or changed by the customer.
3. Products whose trouble cannot be reproduced or identified by YAMAHA.
4. Products utilized, for example, in radiological equipment, biological test equipment applications or for other purposes whose warranty repairs are judged as hazardous by YAMAHA.

THE WARRANTY STATED HEREIN PROVIDED BY YAMAHA ONLY COVERS DEFECTS IN PRODUCTS AND PARTS SOLD BY YAMAHA TO DISTRIBUTORS UNDER THIS AGREEMENT. ANY AND ALL OTHER WARRANTIES OR LIABILITIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE HEREBY EXPRESSLY DISCLAIMED BY YAMAHA. MOREOVER, YAMAHA SHALL NOT BE HELD RESPONSIBLE FOR CONSEQUENTIAL OR INDIRECT DAMAGES IN ANY MANNER RELATING TO THE PRODUCT.

This manual does not serve as a guarantee of any industrial property rights or any other rights and does not grant a license in any form. Please acknowledge that we bear no liability whatsoever for any problems involving industrial property rights which may arise from the contents of this manual.

1. Overview of functions

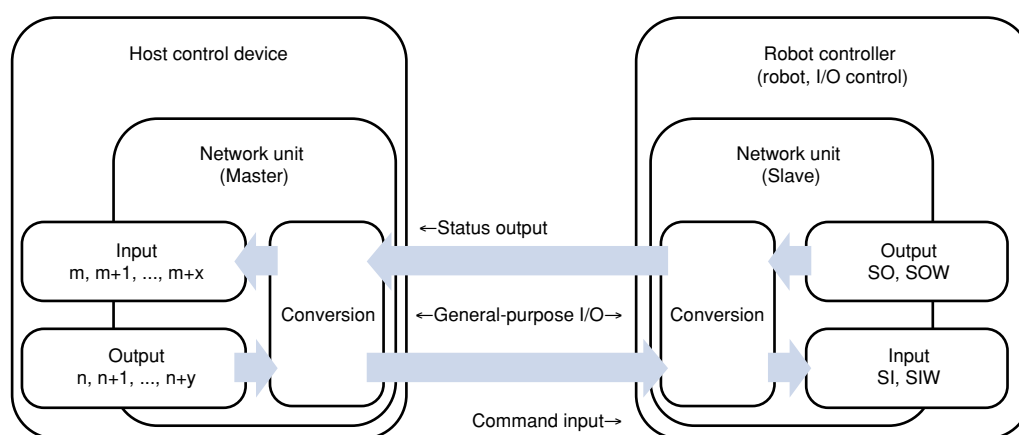
A network interface unit is a device that transmits and receives I/O information via a communication protocol between a controller and a host control device.

We provide a lineup of products that support fieldbus (CC-Link, DeviceNet, PROFIBUS, etc.) or industrial Ethernet (EtherNet/IP, PROFINET, EtherCAT, etc.).

In any of these networks, the robot controller (slave device) operates in accordance with commands via I/O control from a host (master device).

While acting as distributed I/O for the host control device, the network interface unit simultaneously shares I/O information with the controller. The I/O update cycle time between the controller and the network interface unit is 5 ms.

The I/O update cycle time between the network interface unit and the host control device will vary depending on the structure of the applicable network.



I/O consists of a general-purpose input/output area, and a dedicated input/output area that has specific significance such as command input and status output that is optimized for control of robot controllers. Using these, the robot controller can be controlled from the host control device.

The following methods can be used for control via I/O. Combining these methods makes it possible to control the robot.

1. Controlling the robot controller directly using simple commands and status queries via remote I/O's dedicated input/output.

Example) Dedicated input: cancel emergency stop, servo on, reset, start program execution, etc.

Dedicated output: CPU_OK, return-to-origin complete, etc.

2. Using remote commands to control the robot controller directly with advanced instructions and status queries without programming.

Example) < Transmission >

SIW0 = 0x1031 (absolute reset command for robot no.1)

SIW2 = 0x0015 (axis designation: axis 1 through axis 4)

< Reception >

SOW = 0x0100 (executing command)

↓

SOW = 0x0200 (normal end)

3. Using remote I/O's general-purpose input/output to exchange desired information with an external peripheral device, and load it into a robot program or sequence program for execution.

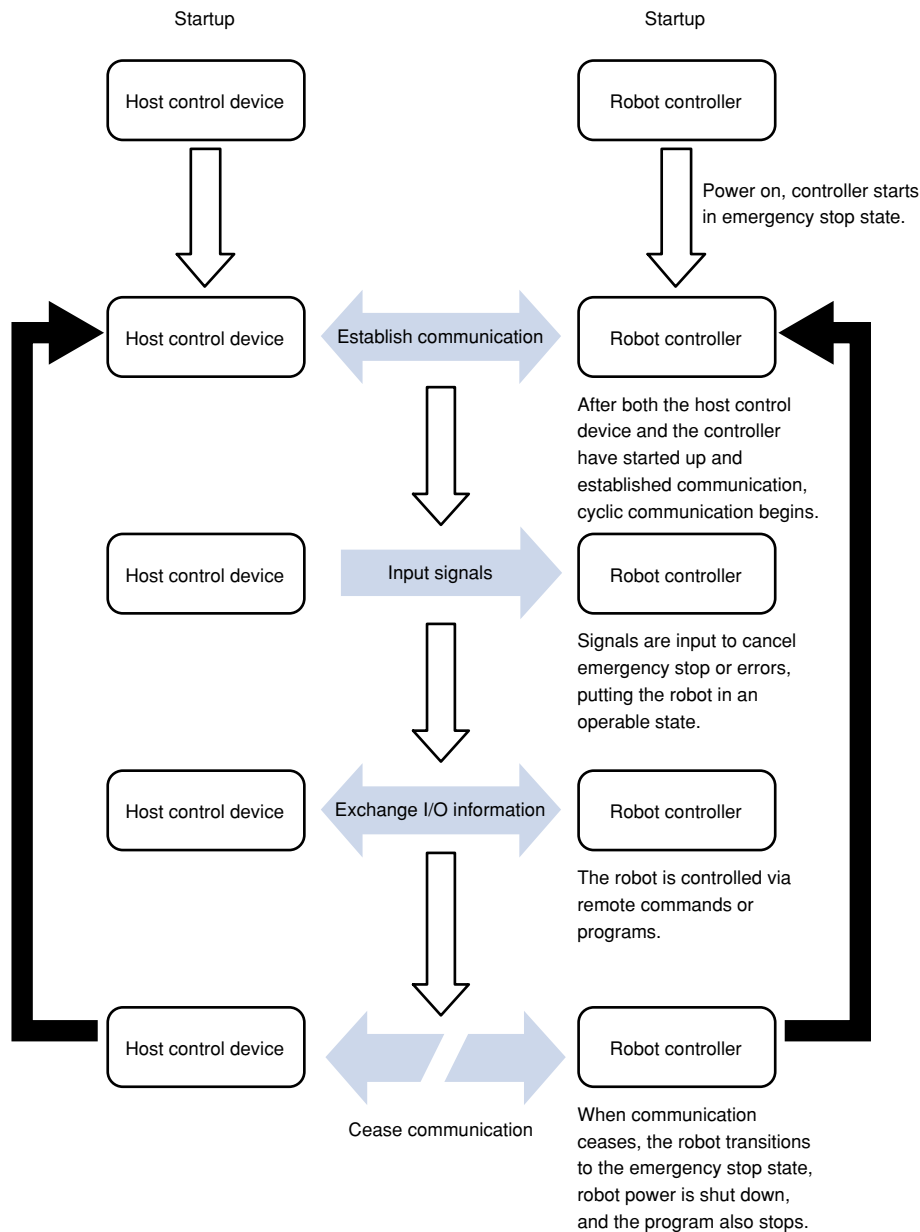
Example) WAIT SI(20)=1, 1000 ; Wait until SI(20) turns on; stop command if it does not turn on after 1 second.

OUT SO2(), 200 ; SO(27--20) turns on, and turns off after 200 ms.

* For details on remote I/O and remote commands, refer to the Remote I/O Manual. For details on robot programs, refer to the RCX340 Programming Manual.

2. Controller status transitions

Here we explain the status transitions of a robot controller equipped with a network interface unit. A controller equipped with a network interface unit will initially start up in the emergency stop state. In order for robot operation to be possible, communication must be established with the host control device, and a sign that cancels the emergency stop state must be input. If communication with the host control device is interrupted for any reason, the controller will again transition to the emergency stop state regardless of the I/O status. In order to resume operation, I/O processing is required, such as reestablishing communication, cancelling the emergency stop state, and resetting any error.



3. Input/output assignments

Here we show the correspondence between the robot controller's serial input/output data and the input/output data on the network.

■ Output from the controller, input to the host control device

Host control device	Controller			
Address (READ)	Address (WRITE)		Input/output type	Use
m	SOW(0)	-	Word output	Command status area
m + 1				Remote command error code area
m + 2	SOW(1)			
m + 3				
m + 4	SOW(2)	SOD(2)	Word output / Double word output	Command response area / General-purpose output area
m + 5				
m + 6	SOW(3)			
m + 7				
m + 8	SOW(4)	SOD(4)		
m + 9				
m + 10	SOW(5)			
m + 11				
m + 12	SOW(6)	SOD(6)		
m + 13				
m + 14	SOW(7)			
m + 15				
m + 16	SOW(8)	SOD(8)		
m + 17				
m + 18	SOW(9)			
m + 19				
m + 20	SOW(10)	SOD(10)		
m + 21				
m + 22	SOW(11)			
m + 23				
m + 24	SOW(12)	SOD(12)		
m + 25				
m + 26	SOW(13)			
m + 27				
m + 28	SOW(14)	SOD(14)		
m + 29				
m + 30	SOW(15)			
m + 31				
m + 32	SO07 - SO00		Bit output	Status output area
m + 33	SO17 - SO10			
m + 34	SO27 - SO20		Bit output / Byte output	General-purpose output area
m + 35	SO37 - SO30			
m + 36	SO47 - SO40			
m + 37	SO57 - SO50			
m + 38	SO67 - SO60			
m + 39	SO77 - SO70			
m + 40	SO107 - SO100			
m + 41	SO117 - SO110			
m + 42	SO127 - SO120			
m + 43	SO137 - SO130			
m + 44	SO147 - SO140			
m + 45	SO157 - SO150			
m + 46	SO167 - SO160			
m + 47	SO177 - SO170			

m: Starting address of the input data area for the target controller

- Each address is 8 bit data.
- SOn() is handled as unsigned 8-bit integer data.
- SOW(n) is handled as unsigned 16-bit integer data.
- SOD(n) is handled as signed 32-bit integer data.
- The upper word and lower word of SOD(n) correspond to SOW(n+1) and SOW(n), respectively.

■ Output from the host control device, input to the controller

Host control device	Controller					
Address (WRITE)	Address (READ)		Input/output type	Use		
n	SIW(0)	-	Word input	Remote command area		
n + 1						
n + 2	SIW(1)			Command data area		
n + 3						
n + 4	SIW(2)	SID(2)	Word input / Double word input	Command data area / General-purpose input area		
n + 5						
n + 6	SIW(3)					
n + 7						
n + 8	SIW(4)	SID(4)				
n + 9						
n + 10	SIW(5)					
n + 11						
n + 12	SIW(6)	SID(6)				
n + 13						
n + 14	SIW(7)					
n + 15						
n + 16	SIW(8)	SID(8)				
n + 17						
n + 18	SIW(9)					
n + 19						
n + 20	SIW(10)	SID(10)				
n + 21						
n + 22	SIW(11)					
n + 23						
n + 24	SIW(12)	SID(12)				
n + 25						
n + 26	SIW(13)					
n + 27						
n + 28	SIW(14)	SID(14)				
n + 29						
n + 30	SIW(15)					
n + 31						
n + 32	SI07 - SI00				Bit input	Command input area
n + 33	SI17 - SI10					
n + 34	SI27 - SI20				But input / Byte input	General-purpose input area
n + 35	SI37 - SI30					
n + 36	SI47 - SI40					
n + 37	SI57 - SI50					
n + 38	SI67 - SI60					
n + 39	SI77 - SI70					
n + 40	SI107 - SI100					
n + 41	SI117 - SI110					
n + 42	SI127 - SI120					
n + 43	SI137 - SI130					
n + 44	SI147 - SI140					
n + 45	SI157 - SI150					
n + 46	SI167 - SI160		Reserved area	Use is prohibited		
n + 47	SI177 - SI170					

n: Starting address of the output data area for the target controller

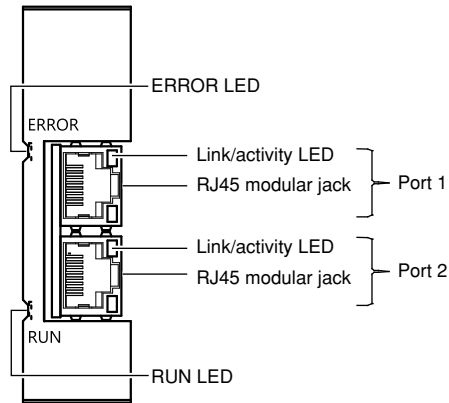
- Each address is 8 bit data.
- SI_n() is handled as unsigned 8-bit integer data.
- SIW_(n) is handled as unsigned 16-bit integer data.
- SID_(n) is handled as signed 32-bit integer data
- The upper word and lower word of SID_(n) correspond to SIW_(n+1) and SIW_(n), respectively.

- * For details on the functions that are assigned to each serial input/output data, refer to the separate Remote I/O Manual.
- * The reserved area cannot be used.

4. Part names and functions

Here we explain the part names and functions of the network interface unit.

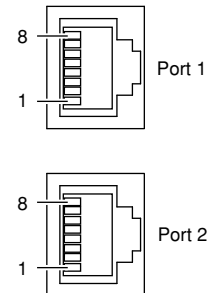
Part names



Connector: RJ45 modular jack

• Pin configuration and connector specifications

Number	Name
1	TXD+
2	TXD-
3	RXD+
4	-
5	-
6	RXD-
7	-
8	-



* An RJ45 modular plug is not included.

ERROR LED

This LED indicates the EtherCAT communication error.

LED status	Explanation
OFF	No error occurs (or power is OFF)
Red flashing	Invalid settings It is disabled to change the status to that sent by the master device as the setting or the register / object is invalid.
Red flashing twice	Application watchdog timeout The sync manager watchdog has timed out.
Red lit*	Application controller malfunction The module is in the "EXCEPTION" status.

RUN LED

This LED indicates the status of the CoE (CANopen over EtherCAT) communication.

LED status	Explanation
OFF	CoE is in the "INIT" status (or power is OFF).
Green lit	CoE is in the "OPERATIONAL" status.
Green flashing	CoE is in the "PRE-OPERATIONAL" status.
Green flashing once	CoE is in the "SAFE-OPERATIONAL" status.
Red lit*	Crucial situation

*: When the either of LED is lit in red, something critical is happening.
The bus interface becomes physically passive in this case.

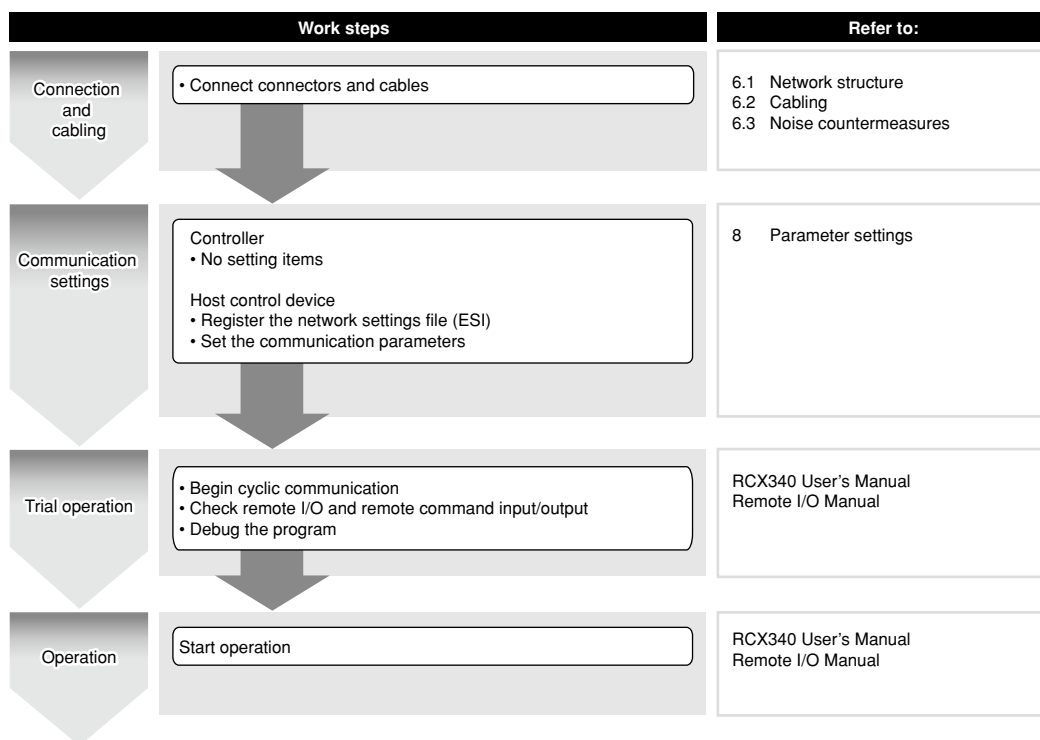
Link/activity LED: Port 1 / Port 2

This LED indicates the link status and activity of EtherCAT.

LED status	Explanation
OFF	A link is not established, and communication is not occurring.
Green lit	An Ethernet link is established, but communication is not occurring.
Green flashing	An Ethernet link is established, and communication is occurring.

5. Flow until operation begins

Here we explain the flow from startup until operation begins. Proceed with the flow shown in the illustration below, and consult the appropriate manuals as necessary for each step.



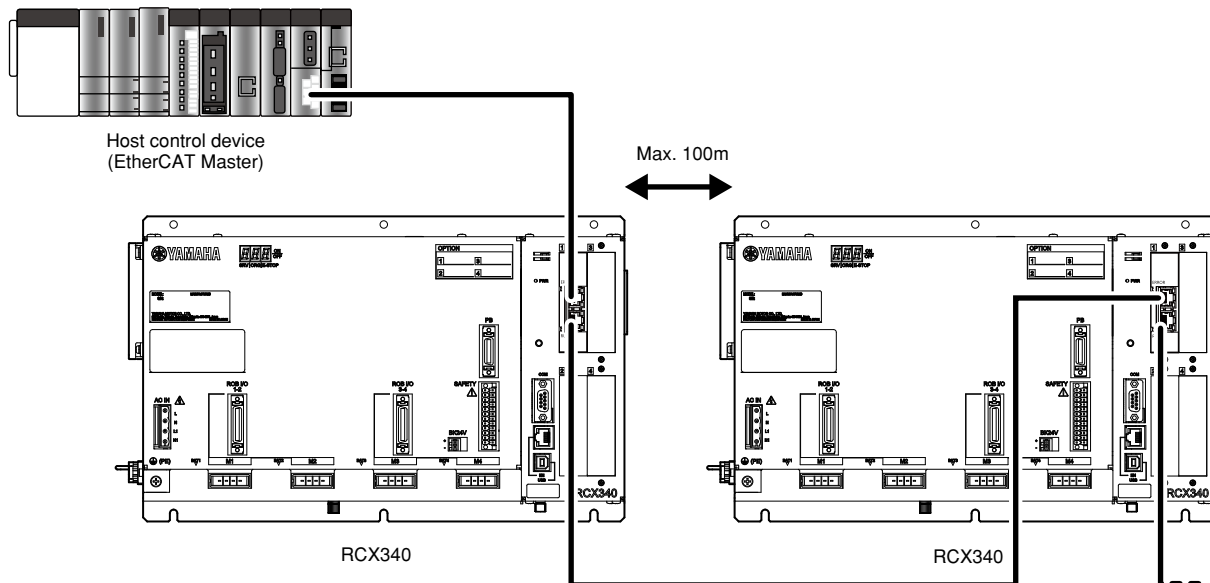
6. Connection

Here we provide an overview of the procedures for connecting the robot controller to the network. Network structure and cable specifications should be designed according to the materials provided by the network interest group (ETG: EtherCAT Technology Group).

6.1 Network structure

EtherCAT is an open network that has been developed by German Beckhoff Automation GmbH and managed by ETG (EtherCAT Technology Group).

The network corresponds to topologies with a lot of flexibility such as Star type, Line type, and Ring type, and it is recommended to construct the network by Daisy-chained connection among them.



CAUTION

Network equipment such as switches can be used if it satisfies the requirements of the EtherCAT network. However if the equipment is to be used in an environment where noise is a concern, such as a factory, equipment that meets EMC industrial standards should be selected.

6.2 Cabling



WARNING

SHUT off the power supply input to the robot controller before performing this work.

Insert the LAN cable's modular plug into the controller's modular jack until you hear it click into place.

■ Cable specifications

Use LAN cables that meet the following specifications.

Category 5e (CAT 5e) or higher

Shielded Twisted Pair (STP: Shielded Twist Pair)

Double-shielded

100 m or shorter (solid-core cable is recommended if it is over 10 m)

(Example selections) SIEMENS AG

Connector plug	IE FC RJ45 Plug 180	6GK1 901-1BB10-2A00
Cable material	IE FC TP standard cable	6XV1 840-2AH10
Tool	Stripping tool for IE FC cable	6GK1 901-1GA00

* OO indicates the number of metal housings (A0: 1 / B0: 10 / E0: 50).

* The cable length can be specified in 1-meter units from a minimum of 20 m to a maximum of 100 m.

6.3 Noise countermeasures

Although the robot controller unit complies with EMC standards and has a certain amount of resistance to external noise, there may be cases in which communication is affected by noise depending on the environment in which the equipment and cable are located.

If communication is affected by noise, a communication error occurs, possibly causing the robot to enter an emergency stop state, or causing throughput to decline. If there is a source of noise, and this noise is suspected of affecting communication, please take the following countermeasures.

1. Equipment and cable selection

As the equipment to be connected to the network, select units that conform to EMC directives. Be sure to use cable of the prescribed specifications.

2. Equipment and cable placement and grounding

If the noise source is nearby, create an appropriate distance between the noise source and the network environment including the cable. For the robot controller unit and equipment connected to it, be sure to attach an appropriate ground as directed by the manual of each unit.

3. Adding anti-noise components

Apply the following measures to the robot controller.

- Install a ferrite core on both ends of the cable immediately adjacent to the connector of the device.
- Install a ferrite core or noise filter on the power supply input of the controller.

7. ESI Files

ESI (EtherCAT Slave Information) is a type of network setting file. It contains settings for a device that supports EtherCAT.

The network setting file contains information necessary for connecting to a host control device, such as I/O size and communication settings, in a prescribed format. When constructing a network, the settings information of all connected network devices must be loaded into the host control device.

The appropriate engineering tool or configuration tool is used to make settings for and write to the host control device, and during this process it will be required to load each device's network setting file into the tools. The steps for loading the file will depend on the host control device to use, and on the tool that supports it. Refer to the manuals for the devices and tools.

These files are included on the robot product manual DVD, and can also be downloaded from the Yamaha product website.

URL: <https://global.yamaha-motor.com/business/robot/download/fieldbus/>
ESI File Name: YAMAHA RCX340 EtherCAT 1_00.xml

8. Parameter Settings

Here we explain the parameters that must be set and the settings files that are required in order to begin communication between the host control device and the robot controller.

Parameters related to EtherCAT

	Name	Setting Range	Initial Value	Description
1	Option board enable	0: INVALID 1: VALID	1: VALID	Enables / disables the option board. Set to VALID in order to use the EtherCAT system. It is required to turn OFF and ON the power after changing the parameter setting.
2	EtherCAT Device ID	0 to 65535	0	Set this parameter to communicate with the host control device using the device ID. 0: No device is set. 1 to 65535: Valid network address



NOTE

It is required to set the device ID or node address (e.g. station alias) for communication between the host device and robot controller.

■ Setting Method



NOTE

Save the data inside the controller to external storage device through support software before changing the controller settings.

Step 1 Open the Parameter Edit screen.

At the initial screen, select (EDIT), press (Enter), and then select (PARAMETER).

Step 2 Press the [F5] (OPTION) key to display the Option Parameters screen.

Step 3 Select the desired parameter.

Use the cursor up/down keys to select the parameter to be edited, then press the (F1) (EDIT) key.

Step 4 Edit the parameter.

Enter the desired set value, then press (Enter).

The "Option board enable" parameter set values display in the order of the option slot numbers.

(1: Upper left → 2: Lower left → 3: Upper right → 4: Lower right)

Specify a setting for the slot number where the EtherCAT compatible module is installed.

Step 5 Press the [ESC] key to end the editing operation.



CAUTION

- Do not touch the conductive areas and the electronic components of the EtherCAT compatible module.
- Do not subject the EtherCAT compatible module to impact shocks.
- Use care to keep moisture and conductive materials away from the EtherCAT compatible module in order to prevent module failure.
- The "12.591: EtherCAT link error" alarm will be displayed at the programming box when the robot controller power is turned ON if there is no connection to the EtherCAT system, or if an EtherCAT system error status exists. Even if this occurs, the setting check described above can be performed.



CAUTION

Data of the robot controller other than network parameters is not passed on automatically.

Be sure to back up from the controller before replacing it, download this data to the controller following replacement, and then connect the controller to the network.

9. Monitoring communication data

On/off data for the host controller device and controller I/O can be monitored via the programming box or the RCX-Studio PC support software. This can be used to check the input/output addresses during setup or for debugging.

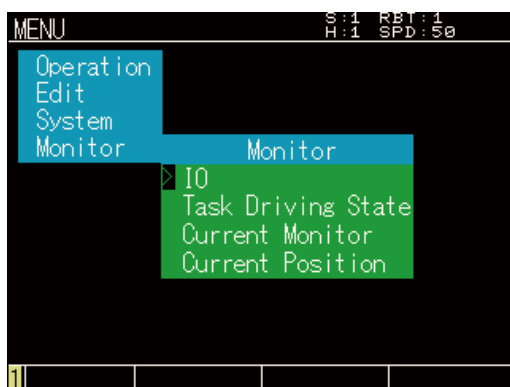


CAUTION

Since the display update interval of the programming box and of RCX-Studio is longer than the data update interval of I/O information, there may be cases in which accurate information is not shown if the on/off interval of the I/O information is short.

9.1 Using the programming box

At the initial screen, select [Monitor]-[I/O] to access the monitor screen.



Valid keys and submenu descriptions in the MANUAL mode are shown below.

Valid keys	Menu	Function
F1	DETAIL	Switches between the input/output list and detail displays.
F2	DI	Displays the DI monitor.
F3	DO	Displays the DO monitor.
F4	MO	Displays the MO monitor.
F5	LO	Displays the LO monitor.
F6	TO	Displays the TO monitor.
F7	SI	Displays the SI monitor.
F7	SO	Displays the SO monitor.
F9	SIW	Displays the SIW monitor.
F10	SOW	Displays the SOW monitor.
MONITOR		Switches the monitor number. It is possible to change the port number display by switching the monitor number. For details on the monitor number, refer to "9.1.1 Switching the port number."



NOTE

Four types of I/O data can be transmitted and received via the network: SI, SO, SIW, and SOW.

9.1.1 Switching the port number

■ Monitor screen

By switching the monitor number the displayed port number can be changed.

The following table shows the correspondence between monitor number and port number.

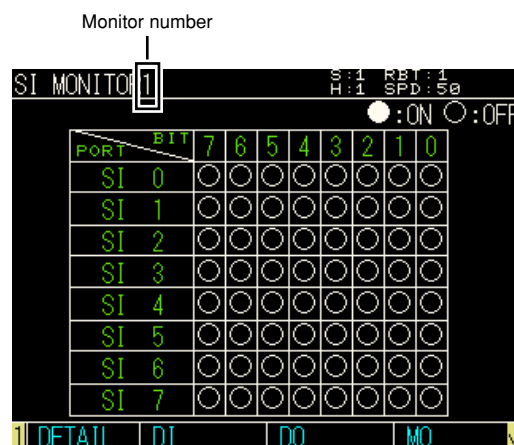
Port name	Monitor number	Display range	Display format
SI,SO	1	☐ 00-77	● : ON ○ : OFF
	2	☐ 100-177	
	3	☐ 200-277	
SIW SOW	1	☐ 0-7	4-digit hexadecimal
	2	☐ 8-15	

* The port name is shown in ☐. For example if "DO monitor 2" (monitor number 2 of the DO port) is shown, DO100-DO177 is shown.

The monitor number is shown in the upper left of the screen.

Press the MONITOR key to switch the monitor number.

■ Monitor number



■ Monitor Detail screen

By switching the monitor number it is possible to change the tens digit in the order of 0 → 10 → 20. The ones digit can be changed by using the cursor keys to select [△] or [▽] and pressing the enter key.

■ Changing the ones digit of the port number



For details on operation, refer to "9.1.3 Input/output detail view."

9.1.2 Input/output list display

Step 1 At the initial screen, select [MONITOR] → [I/O].

The "DI Monitor 1" screen then displays.

Step 2 Select the input/output monitor to be displayed.

Pressing the F2 key (DI)–F10 key (SOW)

displays the input/output monitor

corresponding to the key.

Pressing the MONITOR key changes the port number.

Key	Input/output
F2	DI
F3	DO
F4	MO
F5	LO
F6	TO
F7	SI
F8	SO
F9	SIW
F10	SOW

Press the (ESC) key to end the monitor display.

"SI MONITOR 1" screen

SI MONITOR1

S:1 RBT:1
H:1 SPD:50

●:ON ○:OFF

PORT	BIT	7	6	5	4	3	2	1	0
SI 0		○	○	○	○	○	○	○	○
SI 1		○	○	○	○	○	○	○	○
SI 2		○	○	○	○	○	○	○	○
SI 3		○	○	○	○	○	○	○	○
SI 4		○	○	○	○	○	○	○	○
SI 5		○	○	○	○	○	○	○	○
SI 6		○	○	○	○	○	○	○	○
SI 7		○	○	○	○	○	○	○	○

1|

DETAIL

DI

DO

MO

9.1.3 Input/output details display

Step 1 Open the Input/Output Monitor screen.

Step 2 Press the [F1] (Details) key.

The "Input/Output Details" screen then displays.

Step 3 Changing the port number.

- Changing the tens digit of port number

Pressing the MONITOR key changes the tens digit of the port number in the order of 0 → 10 → 20.

- Changing the ones digit of port number

Use the cursor key to select (Δ) or (▽), then press the (enter) key. The port number can then be changed.

Press the (ESC) key to end the monitor display.

"SI MONITOR 1 DETAIL" screen

SI MONITOR1 DETAIL		S:1	RBT:1
		H:1	SPD:50
		●:ON ○:OFF	
SI 0	▲▼		
Bit	Name	Value	
0		○	
1		○	
2		○	
3		○	
4		○	
5		○	
6		○	
7		○	
1		LIST	DI DO MO

* There are no displays for SIW and SOW details.

9.1.4 Switching the output status

In the "Input/output detail view" screen, DO, MO, LO, TO, and SO output can be switched on/off by individual bits.



CAUTION

- In order to switch the output on/off, the programming box must have control authority.
- SO00-07, SO10-17, and DO00-07 and DO10-17 cannot be changed, since they are dedicated bit outputs that indicate the status of the controller.

Step 1 Display the output port "Monitor" screen.

Press the F3 key (DO)--F6 key (TO) or F8 key (SO) to display the "Monitor" screen.

Step 2 Display the "Detail" screen.

Press the F1 key (Detail) to display the "Detail" screen.

Step 3 Specify the port number.

- Changing the tens digit of port number
Pressing the MONITOR key changes the tens digit of the port number in the order of 0 → 10 → 20.
- Changing the ones digit of port number
Use the cursor key to select (Δ) or (▽), then press the (enter) key. The port number can then be changed.

Step 4 Switch the output status.

Use the cursor keys to select the ON or OFF setting for the bit number which changes the output status, then press the (enter) key to switch that output status.

Press the (ESC) key to end the monitor display.

Example of the output monitor detail screen: "SO MONITOR1 DETAIL"

SO MONITOR1 DETAIL			S:--RBY:1
			H:--SPD:50
SO 2	▲▼	●:ON ○:OFF	
Bit	Name	Value	
0		○ ON OFF	
1		○ ON OFF	
2		○ ON OFF	
3		○ ON OFF	
4		○ ON OFF	
5		○ ON OFF	
6		○ ON OFF	
7		○ ON OFF	
1	LIST	DI	DO MO

9.2 Using RCX-Studio PC support software

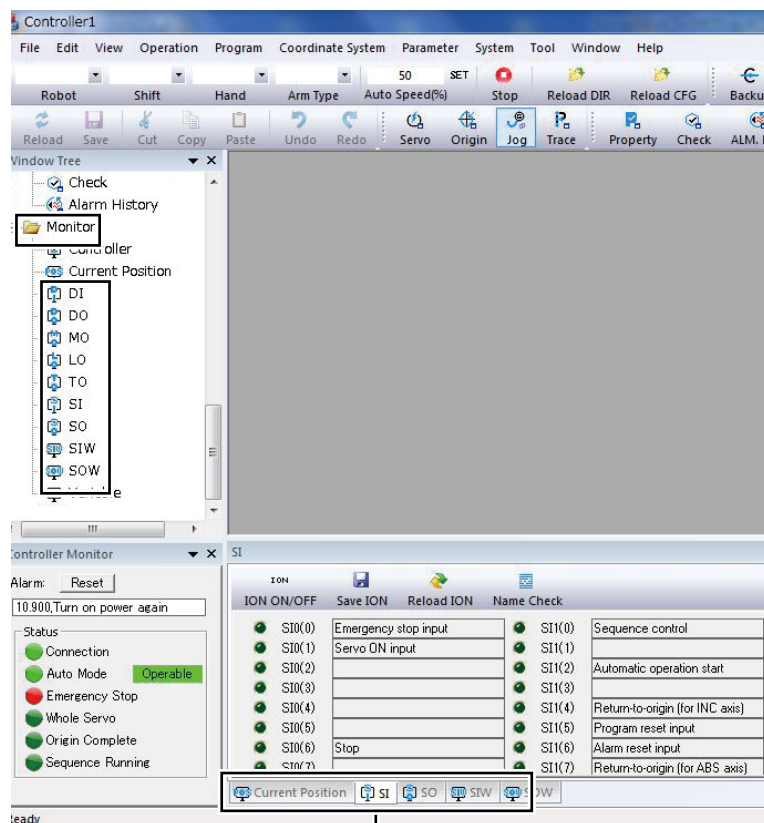
From the window tree, choose [Monitor] and then double-click a port name DI through SOW to monitor, then the monitor screen appears.

The monitor screen contains a list screen and a detail screen. When starting RCX-Studio, the monitor screen that appears first will always be the "Detail screen."

The explanation here uses the example of SI, SO, SIW, and SOW.

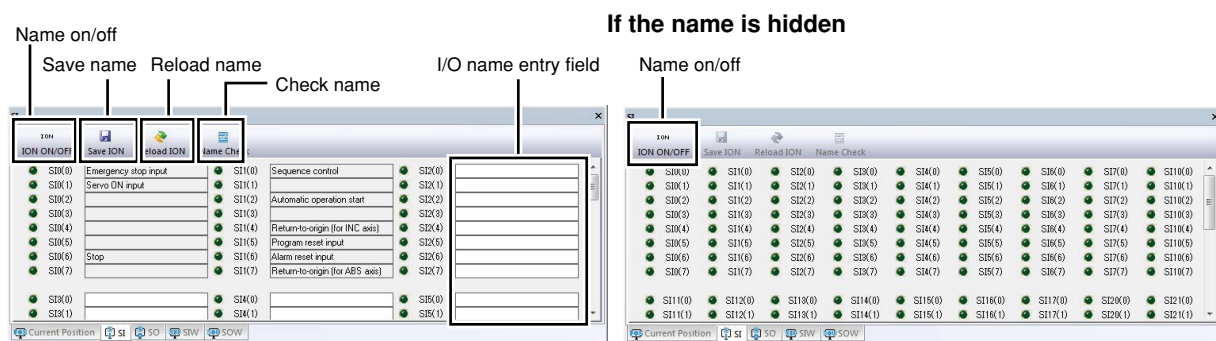
Tip

Once you open a monitor screen, it is shown as a tab at the bottom of the screen. The monitor screen can be displayed by switching tabs.



Tab display

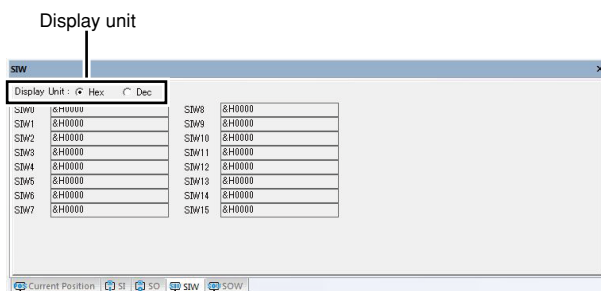
■ SI monitor



	Function
Name on/off	Shows/hides the name.
Save name	Sends the name that is entered in the name entry field to the controller.
Reload name	Loads the name from the controller, updating the indication of the name entry field.
Check name	Checks whether the entered I/O name conforms to the prescribed rules. For details on the naming rules for an I/O name, see "I/O name entry field" below.
I/O name entry field	Enters the I/O name. Observe the following naming rules when entering the name. • The maximum is 16 characters. • Single-byte alphanumeric characters and "_" (underscore) can be used. • Uppercase and lowercase alphabetical characters are distinguished. • A numeral cannot be used as the first character. • It is not possible to assign an I/O name that is identical to a reserved word, or that starts with a system variable name. • It is not possible to assign the same I/O name to more than one I/O. • Since SI00-07 and SI10-17 are dedicated bit inputs, the I/O name cannot be input. Similarly, I/O name input is not possible for DI inputs (DI00-07 and DI10-17).

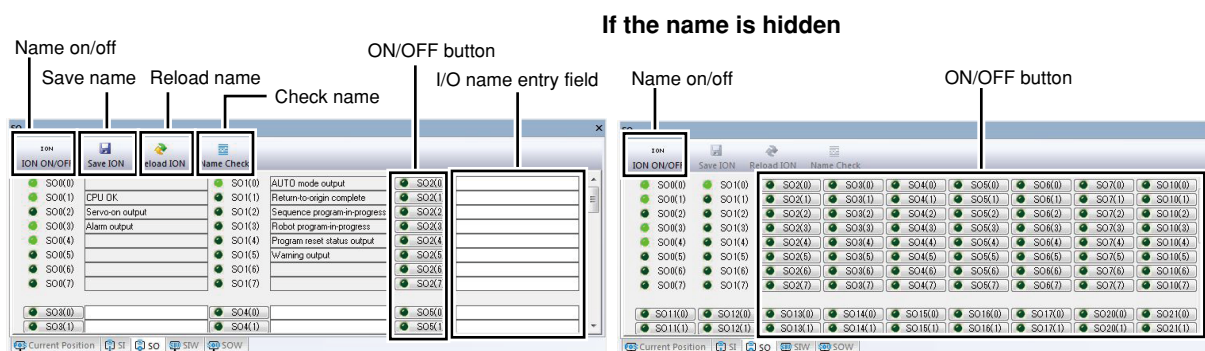
* For a list of reserved words and system variables, and for usage examples of variable names, refer to "13. Reserved word list."

■ SIW monitor



	Function
Display unit	Displays port values in hexadecimal or decimal.

SO monitor



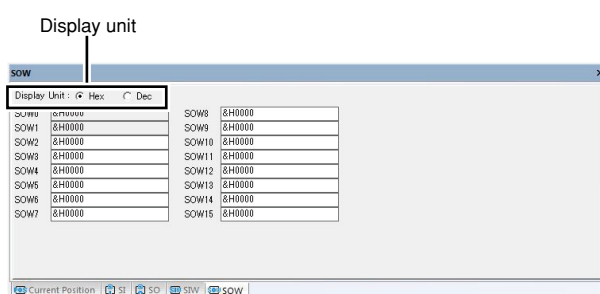
	Function
Name on/off	Shows/hides the name.
Save name	Sends the name that is entered in the name entry field to the controller.
Reload name	Loads the name from the controller, updating the indication of the name entry field.
Check name	Checks whether the entered I/O name conforms to the prescribed rules.
ON/OFF button	Switches the bit output on/off. However since SO00-07 and SO10-17 are dedicated bit outputs that indicate the status of the controller, they cannot be switched on/off. Similarly, DO output (DO00-07 and DO10-17) cannot be switched on/off.
Name entry field	Enters the I/O name. However since SO00-07 and SO10-17 are dedicated bit outputs that indicate the status of the controller, name entry is not possible. Similarly, DO output (DO00-07 and DO10-17) does not allow name entry. For details on the rules for I/O names, refer to "I/O name entry field" in the preceding "SI monitor (Detail screen)."



CAUTION

- In order to switch SO or DO on/off, the programming box must release control authority, and the controller must be in auto mode.
- SO00-07, SO10-17, DO00-07, and DO10-17 are dedicated bit outputs that indicate the status of the controller, and therefore cannot be changed.

SOW monitor



	Function
Display unit	Displays port values in hexadecimal or decimal.

Tip

The SOW2-15 output values can be updated to desired values by rewriting the numerical value in the text box during communication with the controller.

As SOW0 and 1 are dedicated outputs, their values cannot be changed.

As SIW is an external input value, it cannot be changed to an arbitrary value.

10. Disabling an option unit

If the controller is equipped with a field network option unit, it normally communicates with the host control device and controls the robot in accordance with the communicated content, so a ladder program is required for the host control device.

When performing teaching or trial operation of the robot by itself in a state in which the host control device's ladder program has not been provided, the robot can be operated without communicating with the host control device by setting the option board to "disabled".



CAUTION

When shipped from the factory, all functions of the option unit are set to "enabled." In the "enabled" state, the robot first becomes operable when an emergency stop or stop signal is input from the host control device.

For details on I/O functionality, refer to the separate "Remote I/O Manual".

Option board setting parameters

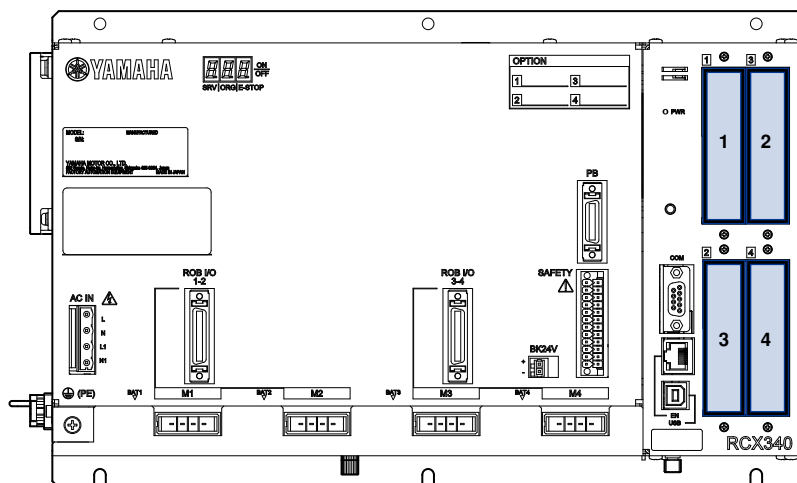
Item		Values	Default value
Option board enabled	Slot 1	0 : INVALID, 1 : VALID	1 : VALID
	Slot 2		
	Slot 3		
	Slot 4		



CAUTION

The "Option board enable" parameter is set individually for option slots 1 through 4. When changing the settings of an option board, be sure to verify the location in which the target option board is installed, and change the setting only for the target slot number.

The illustration below shows the correspondence between the location of each option slot and the actual installation location.



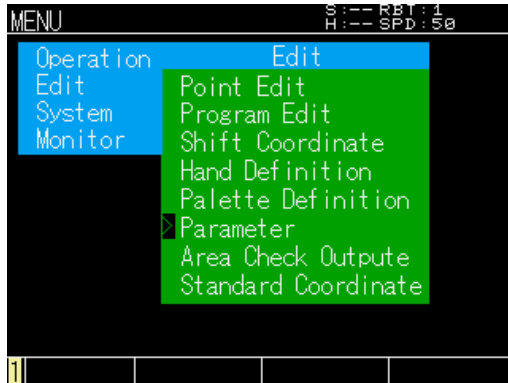
10.1 Making settings

An option board can be disabled either from the programming box or from RCX-Studio PC support software.

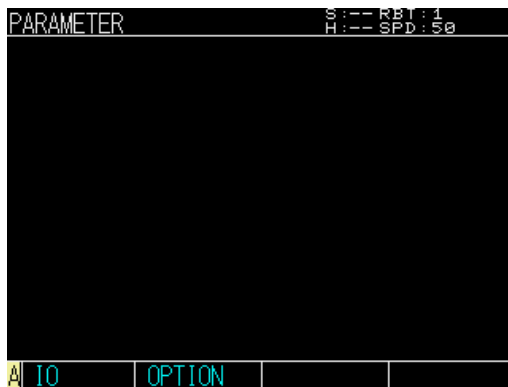
10.1.1 Using the programming box

Step 1 *Open the Parameter Edit screen.*

At the initial screen, select (Edit), press (Enter), and then select (Parameter).



Step 2 *Press the [F6] (OPTION) key to display the Option Parameters screen.*



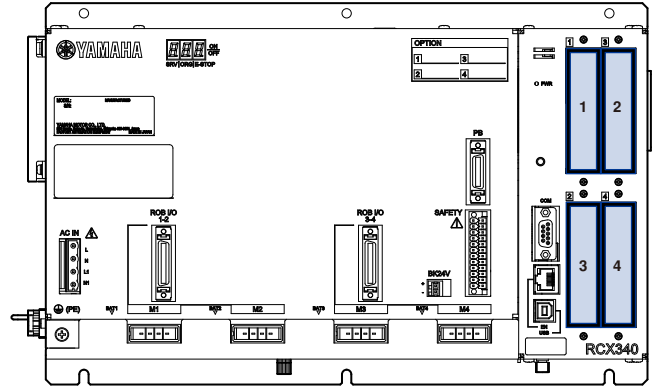
Step 3 *Select the desired parameter.*

Use the cursor up/down keys to select the parameter to be edited, then press the (F1) (EDIT) key.



Step 4 Edit the parameter.

Change the setting for the target slot, and press the enter key.



NOTE

The values of the option board enable parameter correspond to slots 1 through 4 starting from the left. Verify the slot in which the network unit is installed, and change the value only for the target slot number.

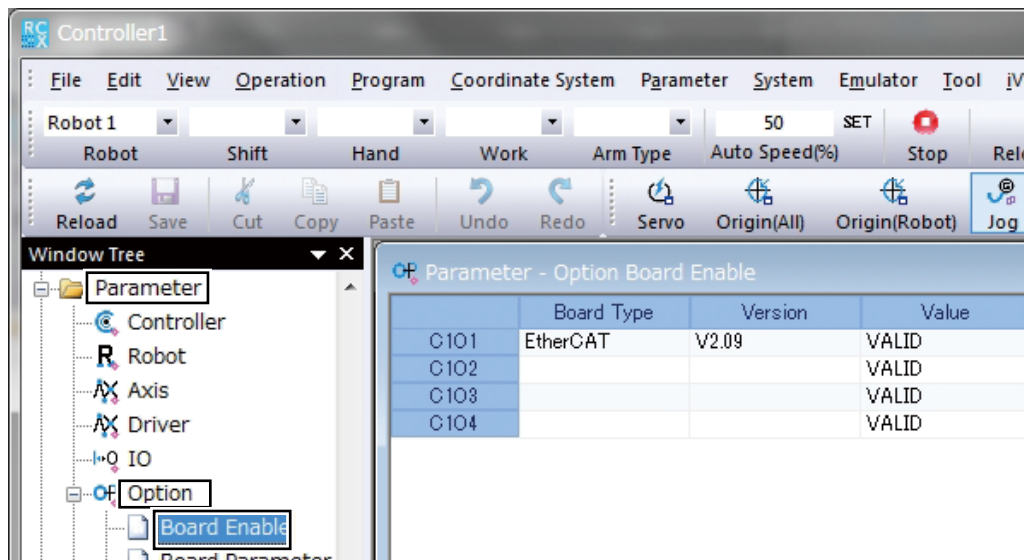
Step 5 Press the [ESC] key to end the editing operation.

The specified content is applied after the controller is restarted.

10.1.2 Using RCX-Studio PC support software

Step 1 In the window tree, select the board parameters.

In the window tree, select (Parameter)-(Option), and double-click (Board Enable).



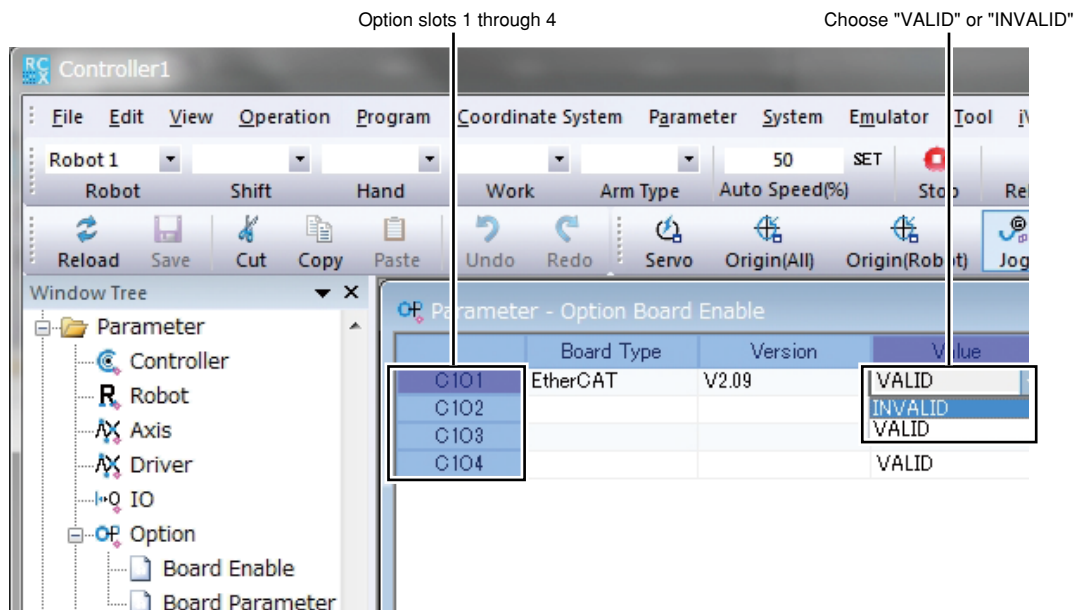
* C101-C104 correspond to slots 1--4.

Verify the slot in which the network unit is installed, and change the value only for the target slot number.

Step 2 Select the option slot "Value" column, and choose either "VALID" or "INVALID."

C101 through C104 in the chart correspond to option slots 1 through 4 of controller 1.

If controllers are linked to each other in a YC-Link/E network, the option slots of controller 2 and following are shown in C201 through C204.

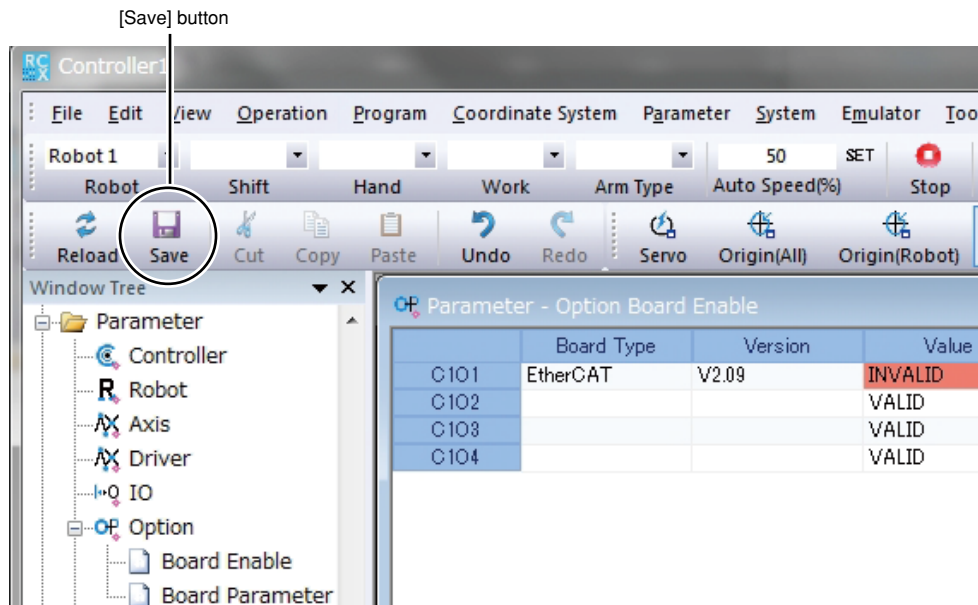
**Tip**

If a cell's value is changed, its background color turns red.

Step 3 Click the [Save] button to save the settings.

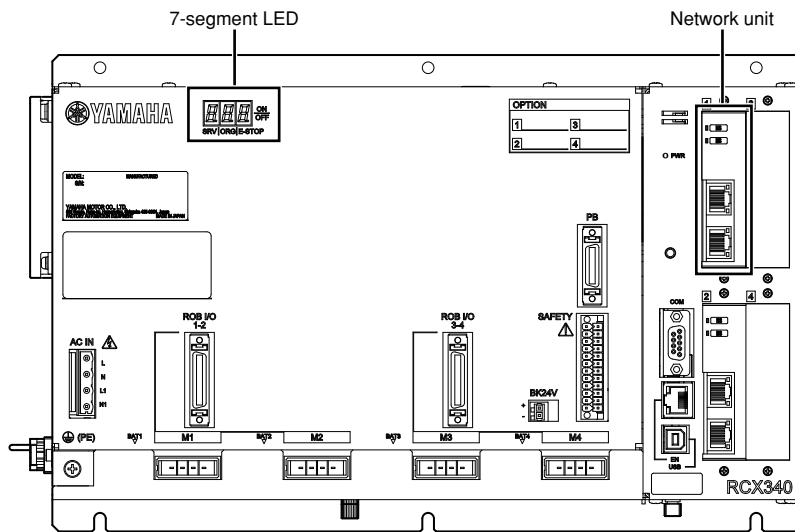
When saving is complete, the cell color changes from red to white.

The specified content is applied after the controller is restarted.



11. Actions to take if communication is not established

If communication is not established, diagnose the situation by referring to the alarm codes or alarm messages of the controller unit and to the network unit's LED illumination pattern, and take the appropriate actions.

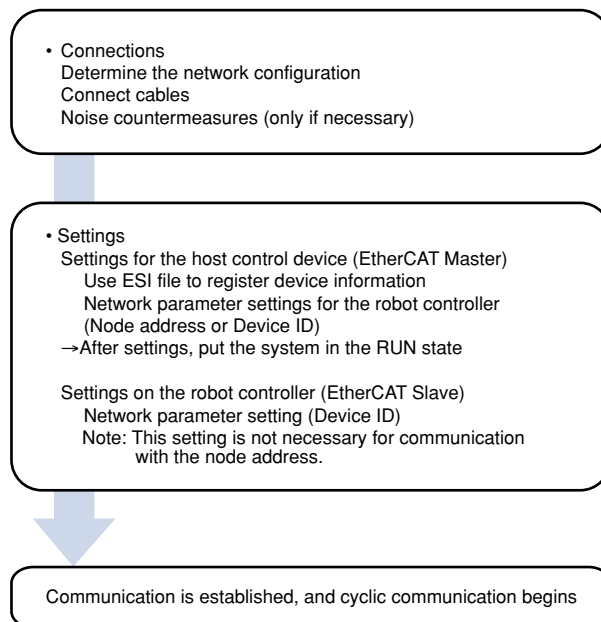


11.1 Reconsider the startup steps

The following steps must be taken when starting up the network system.

If communication cannot be established after the cabling and settings have been completed, it is possible that one of the steps was not performed correctly.

■ Procedure



If communication is not established, check the following points.

- Connections : Do the cables meet the network specifications, and are they securely connected?
- Parameter settings : Are the node address / device ID on the host control device (Master) and the device ID on the controller specified correctly?
- ESI file : Is the correct ESI file registered on the host control device (Master)?

11.2 Viewing the alarm codes in the 7-segment LED

Normal state

- The "PWR" LED on the front of the controller is lit, and the 7-segment LED shows the following.
(Servo off, return-to-origin incomplete, emergency stop cancelled)

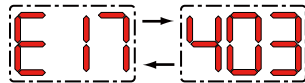


Abnormal state

- The "PWR" LED on the front of the controller is lit, and the 7-segment LED shows an alarm code.
- Note the alarm message on the programming box, and take the actions prescribed by the troubleshooting guidelines.

(Example) Display when an alarm occurs

"E + alarm group number" and "alarm type number" are displayed alternately.



- In the example above, alarm code "17.403" is indicated.
- For details on the alarm messages and content corresponding to each alarm code, refer to "11.3 Diagnostics by alarm message," or the troubleshooting section of the RCX340 controller user's manual.

11.3 Diagnostics by alarm message

Alarm messages related to the network unit are shown below.

Alarm messages not listed in this document may be alarms resulting from the robot controller unit.

Refer to the troubleshooting section of the robot controller user's manual.

11.3.1 Alarms related to network unit connections and settings

These errors occur if communication is not established, or if communication is unstable.

12.591 : EtherCAT link error

Code : &H000C &H024F

Meaning/Cause	a. Error occurred in cable for EtherCAT system. b. The communication setting of the EtherCAT system incorrect. c. The master module power is turned off, or the PLC has stopped operating, or is broken. d. The EtherCAT compatible module is breakdown.
Action	a. Check for a break disconnection, wiring error, short circuit on the EtherCAT cable or the specifications (cable length, etc.). b. Check the communication setting. c. Check that the master module operates correctly. d. Replace the EtherCAT compatible module.

12.592 : EtherCAT overtime error

Code : &H000C &H0250

Meaning/Cause	a. Communication error occurred by noise, etc. in the EtherCAT system. b. Master module power is turned OFF or has stopped operating. c. The cable is broken or unconnected.
Action	a. Take the noise preventive actions for the cable and controller of the EtherCAT system. b. Check that the master module operates correctly. c. Check the EtherCAT cable connectio

11.3.2 Alarms related to I/O on/off status

These errors occur depending on the state of the dedicated input/output.

If the robot controller is equipped with a network unit, the robot cannot be operated unless an emergency stop or STOP signal is input from the host control device, even if the robot is being controlled from somewhere other than the host control device.

For details on I/O functions, refer to "Remote I/O Manual."

12.400 : Stop input on

Code : &H000C &H0190

Meaning/Cause	a. Program execution or axis movement was attempted in the stop status. b. The robot was put in the stop status during program execution or axis movement. c. 24V-power for I/O is not supplied to the DIO connector. d. The DIO connector is not connected.
Action	a, b. Cancel the stop status, and then execute the program or move the axis. c. Supply 24V-power for I/O. d. Connect the DIO connector. * Set the "Option board enable" parameter INVALID when DIO is not used.

12.600 : Emergency stop on

Code : &H000C &H0258

Meaning/Cause	a. The programming box emergency stop button was pressed. b. The emergency stop terminal on the SAFETY connector is open (emergency stop status). c. The programming box or terminator is not connected to the PB connector. d. The SAFETY connector is not connected.
Action	a. Release the emergency stop button on the programming box. b. Close the emergency stop terminal on SAFETY connector. c. Connect the programming box or terminator to the PB connector. d. Attach the SAFETY connector.

11.3.3 Alarms that the customer cannot resolve

These are error messages that are not expected to occur in the field.

If one of these errors occurs, please contact your distributor with details of the situation.

12.767 : EtherCAT initialize error

Code : &H000C &H02FF

Meaning/Cause	Initializing the EtherCAT option board failed.
Action	Contact your distributor.

12.900 : Incorrect option setting

Code : &H000C &H0384

Meaning/Cause	a. Error occurred in ID setting on the option module. b. Option modules that cannot be mixed were installed. c. The installed option module cannot be identified.
Action	a. Check the ID setting of the option module. b. Install the correct option modules. c. Replace the option module. • Replace the controller.

12.904 : SIO option board initialize error

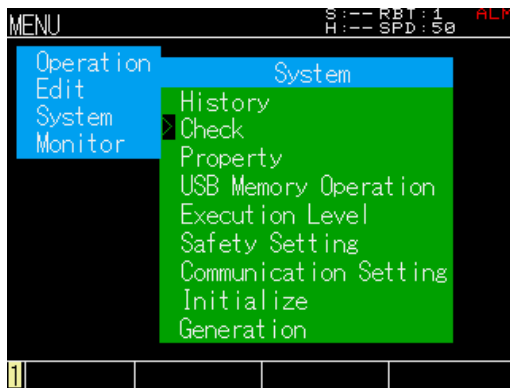
Code : &H000C &H0338

Meaning/Cause	Initializing the SIO option board failed.
Action	Contact your distributor.

11.3.4 Checking alarm codes in the programming box

Step 1 Open the *Parameter Edit* screen.

At the initial screen, select (Edit), press (Enter), then select (Parameter).



Step 2 Check the alarm code that is occurring.

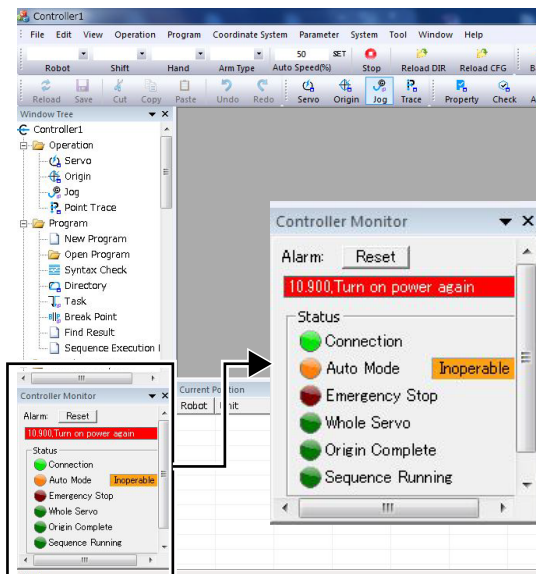
Press the F1 key (RELOAD) to update the display to the current state.

In the illustration, alarm code 10.900 is occurring.



11.3.5 Checking alarm codes in RCX-Studio PC support software

Alarm codes and alarm messages are shown in the "Controller Monitor" located in the lower left of the screen.



12. Specifications

12.1 Network specifications

Spec. Item	Contents	
Supported software versions	RCX340	: V1.62 or later
	PBX/PBX-E	: V1.08 or later
	RCX-Studio Pro	: V2.1.9 or later
ESI file name	YAMAHA RCX340 EtherCAT 1_00.xml	
Transmission speed	100Mbps (Auto-negotiation)	
Connector specifications	RJ-45 connector (8-pole modular connector) 2 ports	
Conforming cable specifications	CAT 5e or higher STP cable (double shield)	
Max. cable length	100 m	
Monitor LEDs	RUN, ERROR, Link/Activity:Port1-2	
Input/output data size	Input : 48bytes	Dedicated word input 2 words (4 bytes)
		General-purpose word input 14 words (28 bytes)
		Dedicated bit input 16 bits (2 bytes)
		General-purpose bit input 112 bits (14 bytes)
	Output : 48bytes	Dedicated word output 2 words (4 bytes)
		General-purpose word output 14 words (28 bytes)
		Dedicated bit output 16 bits (2 bytes)
		General-purpose bit output 112 bits (14 bytes)

12.2 Input/output specifications

Controller ⇒ Host control device			Host control device ⇒ Controller		
Address (WRITE)		I/O function	Address (READ)		I/O function
SOW(0)	-	Command status area	SIW(0)	-	Remote command area
SOW(1)		Command response area	SIW(1)		Command data area
SOW(2)	SOD(2)	Command response area / General-purpose output area	SIW(2)	SID(2)	Command data area / General-purpose input area
SOW(3)			SIW(3)		
SOW(4)	SOD(4)		SIW(4)	SID(4)	
SOW(5)			SIW(5)		
SOW(6)	SOD(6)		SIW(6)	SID(6)	
SOW(7)			SIW(7)		
SOW(8)	SOD(8)		SIW(8)	SID(8)	
SOW(9)			SIW(9)		
SOW(10)	SOD(10)		SIW(10)	SID(10)	
SOW(11)			SIW(11)		
SOW(12)	SOD(12)		SIW(12)	SID(12)	
SOW(13)			SIW(13)		
SOW(14)	SOD(14)		SIW(14)	SID(14)	
SOW(15)			SIW(15)		
SO07 - SO00	SO00	Emergency stop status output	SI07 - SI00	SI00	Emergency stop input
	SO01	CPU_OK status output		SI01	Servo on input
	SO02	Servo on status output		SI02	Reserved area (do not use)
	SO03	Alarm status output		SI03	
	SO04	MPRDY status output		SI04	
	SO05	Reserved area (do not use)		SI05	
	SO06			SI06	Stop input
	SO07			SI07	Reserved area (do not use)
SO17 - SO10	SO10	Auto mode status output	SI17 - SI10	SI10	Sequence control input
	SO11	Return-to-origin complete status output		SI11	Reserved area (do not use)
	SO12	Sequence program execution status output		SI12	Automatic operation start
	SO13	Robot program operating output		SI13	Reserved area (do not use)
	SO14	Program reset status output		SI14	Return-to-origin input (incremental type axis)
	SO15	Warning output		SI15	Program reset input
	SO16	Reserved area (do not use)		SI16	Alarm reset input
	SO17			SI17	Return-to-origin input (absolute type axis)
SO27 - SO20		General-purpose output area	SI27 - SI20		General-purpose input area
SO37 - SO30			SI37 - SI30		
SO47 - SO40			SI47 - SI40		
SO57 - SO50			SI57 - SI50		
SO67 - SO60			SI67 - SI60		
SO77 - SO70			SI77 - SI70		
SO107 - SO100			SI107 - SI100		
SO117 - SO110			SI117 - SI110		
SO127 - SO120			SI127 - SI120		
SO137 - SO130			SI137 - SI130		
SO147 - SO140			SI147 - SI140		
SO157 - SO150			SI157 - SI150		
SO167 - SO160		Reserved area (do not use)	SI167 - SI160		Reserved area (do not use)
SO177 - SO170			SI177 - SI170		

For details on each I/O function, refer to the separate "Remote I/O Manual".

13. Reserved word list

Because the following names are reserved for the robot language, they cannot be used as I/O names.

A	DEF	IDIST	N
ABS	DEGRAD	IF	NAME
ABSADJ	DELAY	IMP	NEXT
ABSRPOS	DI	INCH	NOT
ACC	DIM	INCHT	O
ACCEL	DIR	INCHXY	OFF
ACCESS	DIST	INIT	OFFLINE
ACO	DO	INPUT	ON
ALL	DPM	INT	ONLINE
ALM	DRIVE	ION	OPEN
ALMRST	DRIVEI	J	OPT
AND	DRV	JL	OR
ARCHP1	DS	JOG	ORD
ARCHP2	DSPEED	JOGT	ORGORD
ARM	E	JOGXY	ORGRTN
ARMCND	ELSE	JTOXY	ORIGIN
ARMSEL	ELSEIF	L	OUT
ARMTYP	EMG	LEFT	OUTPOS
ARP	END	LEFTY	P
ARY	ENDIF	LEN	P
ASPEED	EOF	LET	PATH
ATN	EQV	LINEMODE	PC
ATN2	ERA	LOAD	PCM
ATTR	ERL	LOC1	PDEF
AXWGHT	ERR	LOC2	PGM
B	ERROR	LOC3	PGMTSK
BIN	ETH	LOC4	PGN
BREAK	ETHSTS	LOC5	PLS
C	EXIT	LOC6	PLT
CALL	EXITTASK	LOCF	PMOVE
CASE	F	LOG	PNM
CFG	FN	LSHIFT	PNT
CHANGE	FOR	M	PPNT
CHGPRI	FREE	MAINPG	PRINT
CHR	G	MCHREF	PRM
CLOSE	GEP	MEM	PSHFRC
CMU	GEPSTS	MID	PSHJGSP
CNT	GO	MNS	PSHMTD
CONT	GOSUB	MOD	PSHRSLT
COPY	GOTO	MODE	PSHSPD
COS	H	MOTOR	PSHTIME
CURTQST	HALT	MOVE	PUSH
CURTRQ	HALTALL	MOVEI	PWR
CUT	HAND	MOVET	R
D	HEX	MRF	RADDEG
DATE	HND	MRKSET	RBT
DBP	HOLD	MSG	READ
DEC	HOLDALL	MSGCLR	REF
DECEL	I	MSPEED	REM

REN	SGI	SWI	VEL
RESET	SGR	SYNCHK	VER
RESTART	SHARED	T	W
RESUME	SHIFT	TAG	WAIT
RETURN	SI	TAN	WEIGHT
RIGHT	SID	TASKS	WEND
RIGHTY	SIN	TCHXY	WHERE
RSHIFT	SIW	TCOUNTER	WHILE
RUN	SKIP	TEACH	WHRXY
RUNTO	SKIPTO	THEN	WHRXYEX
S	SO	TIM	WRITE
S	SOD	TIME	X
SCK	SOW	TIMER	XOR
SELECT	SPEED	TO	XY
SEND	SQR	TOLE	XYTOJ
SEQCMPL	START	TORQUE	Y
SEQUENCE	STEP	TSKECD	YZ
SERVO	STOP	TSKMON	Z
SET	STOPON	TSKPGM	ZX
SETGEP	STR	V	
SETPW	SUB	VAL	
SFT	SUSPEND	VAR	

List of system variables

Because the following names are used as system variable names, they cannot be used at the beginning of other variable names (n: numeric value).

A	FN	MO _n	Sn
Ac _n	G	P	Son
D	GP _n	PC _n	SONM _n
DIn	H	P _n	T
DINM _n	H _n	PN _n	TO _n
DO _n	L	S	
DONM _n	LO _n	SIn	
F	M	SINM _n	

Variable name usage examples

- Although keywords which are reserved as robot language words cannot be used as they are, **they can be used as variable names if alphanumeric characters are added to them.**

Example: "ABS" cannot be used, but "ABS1" or "ABSX" can be used.

- Keywords reserved as system variables **cannot be used at the beginning of other variable names**, even if alphanumeric characters are added to them.

Example: "FN" cannot be used. "FNA" and "FN123" also cannot be used.

Revision Record

Manual version	Issue date	Description
Ver. 1.00	Aug. 2018	First edition
Ver. 1.01	Oct. 2018	Clerical error corrections, etc.

User's Manual

Network Board RCX340

EtherCAT

Oct. 2018

Ver. 1.01

YAMAHA MOTOR CO., LTD. Robotics Operations

All rights reserved. No part of this publication may be reproduced in any form without the permission of YAMAHA MOTOR CO., LTD.
The contents of this manual are subject to change without prior notice.
Information furnished by YAMAHA in this manual is believed to be reliable. However, no responsibility is assumed for possible inaccuracies or omissions. If you find any part unclear in this manual, please contact your distributor.

Robotics Operations

127 Toyooka, Kitaku, Hamamatsu, Shizuoka, 433-8103, Japan
Tel. 81-53-525-8250 Fax. 81-53-525-8378

Robot manuals can be downloaded from our company website.
Please use the following for more detailed information.

<https://global.yamaha-motor.com/business/robot/>



YAMAHA

YAMAHA MOTOR CO., LTD.